

Examining Economic Irrationality with Neuroeconomics

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Agents in an economy are expected to act rationally by maximizing their utility, as suggested by economic models. Irrationality thus arises from the expectation of perfect choice-making behavior needed for prediction. This phenomenon of irrational monetary decision-making led to the field of behavioral economics, combining behavioral insights from psychology with a monetary medium provided by economics (Cohen, 2010). Although behavioral economics provided major insights into such obscure phenomena, it lacked mechanistic explanatory power. Standard economic models frequently operate under the principle of *ceteris paribus* assumptions, holding constant many psychological and biological influences in order to generate clean predictions about behavior. However, when these “held constant” factors systematically shape decision-making, reduction to a lower-level science, such as neuroscience, can be useful for identifying the underlying mechanisms that produce deviations from rational choice. Neuroeconomics attempts to fill that gap by integrating neuroscience’s experimental methods, such as brain imaging (fMRI), electrophysiology, and lesion studies. By directly observing neural activity during decision-making tasks, researchers can identify the biological processes that shape preferences, risk perception, reward evaluation, and self-control, allowing economists to move beyond purely behavioral descriptions and begin linking deviations from rational choice to specific neural circuits and cognitive systems.

It has been theorized that economic decisions emerge through competing neural systems. Rangel et al. (2008) suggest that there are three types of valuation systems: Pavlovian, habitual, and goal-directed. While they share similar processes and even biological bases, these systems are potentially responsible for many irrational economic phenomena such as loss aversion and ambiguity aversion. Economic irrationality, defined as non-utility-maximizing choice, is therefore a predictable result of neural processing best modeled by the Pavlovian-habitual-goal-directed systems.

Framework

Pavlovian behaviors are automatic responses triggered by environmental stimuli associated with reward or punishment (Rangel, 2008).. The neural architecture of the Pavlovian system begins in the amygdala, where emotional value is encoded; signals are then sent to different parts of the brain to trigger appropriate subconscious behaviors (Dayan et al., 2008). Signals sent to the striatum are associated with the key function of decision-making. The dorsal striatum automates behavior in habit formation, where dopamine signaling over time creates strong neural connections driving unconscious decisions. The ventral striatum encodes reward prediction error; dopamine spikes when expectations are exceeded and falls when expectations fail (Rangel, 2008). Signals may also be sent to the periaqueductal gray, which is in part responsible for freezing and defense, a component of the automatic fight-or-flight response (Rangel, 2008). Additionally, the insular cortex plays a role in aversion, linking emotional reactions to external stimuli (Paulus & Stein, 2006). Knutson et al. (2007) provide an example of this process in an economic context, showing that activation in the nucleus accumbens, a key component of the ventral striatum, predicts purchasing decisions before individuals consciously

report wanting the product. Heightened NAcc activity in response to anticipated rewards was associated with an increased likelihood of choosing to buy, adding merit to the idea of Pavlovian systems influencing decisions. Pavlovian actions can be classified as somewhat hard-wired; the system acts immediately in response to external stimuli and is not altered by prediction error learning. The Pavlovian system can therefore produce irrational actions in the absence of regulation by conscious processes.

The habitual system is similar to the Pavlovian system in that both generate quick and automatic responses, but they are distinct in a critical way. The habitual system elicits responses that have been learned through repeated environmental interactions rather than biologically ingrained mechanisms for survival. The habitual system builds stimulus–response associations through trial and error (Rangel, 2008). The value of an action is assigned based on the expected reward from similarly experienced situations. This model relies on dopaminergic prediction error learning, where dopamine signals whether outcomes are better or worse than expected, strengthening or weakening behaviors over time (Camerer, 2005). Cognitively, habit control is centered in the dorsolateral striatum and is supported by cortico–basal ganglia–thalamic loops, feedback circuits that guide action, behavior, and learning. Evidence for this system comes from Yin and Knowlton (2006), where lesions in the dorsolateral striatum shift what would normally be habitual behavior back to conscious control.

Cognitively, habit control is centered in the dorsolateral striatum and is supported by cortico-thalamic loops, a feedback circuit often called the “decision circuit” that guides action, behavior, and learning. The cortico-thalamic loop has the potential to automate behavior through repeated signaling. Evidence for this system comes from Yin and Knowlton (2006), where

lesions in the dorsolateral striatum shift what would normally be habitual behavior back to conscious control. These structures store associations that become stable, fast, and automatic. With enough reinforcement, the cortico-thalamic loop can execute actions without conscious processing (Rangel, 2008). When cognitive processes are overridden, habitual systems can create predictable biases in an economic context.

The goal-directed valuation system supports flexible and deliberate decision-making by computing value through explicit action–outcome–reward representations derived from prior knowledge or experience. Unlike the other systems, the goal-directed process uses top-down processing to create behaviors that maximize future payoff (Rangel, 2008). Although little is known about the internal representations of action–outcome and action–reward associations, several studies suggest that the orbitofrontal cortex and the dorsolateral prefrontal cortex—areas associated with higher-level control—integrate outcome value and support rational behavior (Rangel, 2008). Research in neuroeconomics also suggests that these valuation systems converge on a common neural value signal, often associated with activity in the ventromedial prefrontal cortex, allowing the brain to compare and choose between competing options. Because the goal-directed system requires greater cognitive resources, not all decisions are processed through it (Camerer, 2005).

Several studies also highlight the importance of the associative cortico-basal ganglia loop, which processes and integrates information to support goal-directed action. Goal-directed system valuations are most strongly associated with rational economic behavior, passing through higher-order cognitive systems like the orbitofrontal cortex. However, because of the cognitive resources required, not all decisions pass through this primary system (Camerer, 2005).

Examples

Ambiguity aversion occurs when an agent prefers known options over ambiguous or hidden ones (Pushkarskaya et al., 2015). Ambiguity aversion can be classified as a byproduct of the Pavlovian and habitual systems; ambiguous options provoke feelings of discomfort by activating areas associated with emotional responses (Pushkarskaya et al., 2015). The anterior insula, in particular, activates feelings of aversion, while the amygdala activates when ambiguity increases, producing threat-based reactions that push decisions toward well-defined options (Hsu et al., 2005).

Ambiguity aversion is speculated to have an evolutionary basis, where a preference for what is known stems from perceived safety in certainty. The Pavlovian system primarily influences actions when the goal-directed system is overloaded or unconsciously bypassed (Rangel et al., 2008), leading to nonoptimal decision-making in economic contexts. This reflects a hard-wired neurological process rather than deliberate irrational choice.

These neural mechanisms hold explanatory power over decisions often thought of as “irrational” according to economic principles. Loss aversion occurs when an agent behaves inconsistently across a series of gambles. Typically, this preference manifests as a tendency to weight losses more heavily than equivalent gains (Tversky & Kahneman, 1991). In behavioral economics, this phenomenon is often viewed as a cognitive bias, but integrating neuroeconomics reveals that it may arise from asymmetric neural responses.

This system produces encoding asymmetry in reactions, meaning losses generate stronger neural responses than equivalent wins (Rangel, 2008). The amygdala, striatum, and ventromedial

prefrontal cortex exhibit larger decreases in activity during losses compared to activity spikes from equivalent gains (Fox et al., 2007). Consistent negative signal reinforcement elicits strong activation of the habitual system to avoid loss. Similar to other systems, the habitual system can only strongly influence actions when cognitive control is limited, meaning that loss aversion can sometimes be reduced through deliberate and slow reasoning (Fox et al., 2007).

Criticisms

As with most neuroscience claims, evidence remains correlational rather than causal. Lesion and fMRI activation studies that examine the basis of “economic irrationality” can only identify correlations between neural activity and economic decisions. They cannot establish causal relations because studies primarily observe patterns of brain activation alongside behavior without experimentally manipulating neural structures directly. Ambiguity aversion or loss aversion may also arise from cultural and environmental factors that neuroeconomics alone cannot fully capture.

A key limitation of the neuroeconomics approach is that neural mechanistic explanations do not capture the primary domain of economics and therefore cannot be used as a major critique of economic theory. Gul and Pesendorfer (2005) argue that standard economic models are not intended to capture processes of the brain but are designed to predict choice behavior using preference data. Because traditional models do not make claims regarding underlying physiology, evidence of neural processes cannot directly refute them.

From Gul and Pesendorfer’s account, describing economic phenomena like loss aversion in terms of neural mechanisms may provide a richer account. However, it does not automatically

improve the explanatory power of economics unless it generates new, testable behavioral predictions beyond what can already be represented in standard choice-based frameworks. The purpose of introducing neuroeconomic evidence in this paper is not to refute standard economic theory but to bring attention to a burgeoning field within the economic discipline that has the potential to provide biological bases for economic behavior. While choice-based error models remain powerful predictive tools, neuroeconomics helps explain why certain deviations from standard assumptions occur systematically. This additional layer adds important context because it identifies biological mechanisms that may reveal when “irrational” phenomena occur.

Conclusion

What classical economic theory calls “irrational” decision-making may be better understood as natural processes designed to support survival and automate decisions. Although what might be beneficial for cognitive efficiency may result in non-utility-maximizing choices, there is reason to believe that aversions to loss, ambiguity, and other behavioral biases arise from intrinsic mechanisms designed to distribute scarce cognitive resources in times of need. Neuroeconomics offers explanations for these deviations that do not rely solely on assumptions of reasoning failure but instead on predictable products of neural architecture

References

- Camerer, C., Loewenstein, G., & Prelec, D. (2005). Neuroeconomics: How neuroscience can inform economics. *Journal of Economic Literature*, 43(1), 9–64. <https://doi.org/10.1257/0022051053737843>
- Clark, C. (2024, December 15). *Why we buy what we buy: The neuroscience of shopping*. Neuroscience News. <https://neurosciencenews.com/neuroeconomics-shopping-neuroscience-28247/>
- Cohen, J. (2010, January 15). *What is neuroeconomics?* Yale Insights. <https://insights.som.yale.edu/insights/what-is-neuroeconomics>
- Dayan, P., & Seymour, B. (2008). Values and actions in aversion. In P. W. Glimcher, C. F. Camerer, E. Fehr, & R. A. Poldrack (Eds.), *Neuroeconomics: Decision making and the brain* (pp. 175–191). Elsevier.
- Graybiel, A. M. (2008). Habits, rituals, and the evaluative brain. *Annual Review of Neuroscience*, 31, 359–387. <https://doi.org/10.1146/annurev.neuro.29.051605.112851>
- Gul, F., & Pesendorfer, W. (2005). *The case for mindless economics*. Princeton University.
- Hsu, M., Bhatt, M., Adolphs, R., Tranel, D., & Camerer, C. F. (2005). Neural systems responding to degrees of uncertainty in human decision-making. *Science (New York, N.Y.)*, 310(5754), 1680–1683. <https://doi.org/10.1126/science.1115327>
- Knutson, B., Rick, S., Wimmer, G. E., Prelec, D., & Loewenstein, G. (2007). Neural predictors of purchases. *Neuron*, 53(1), 147–156. <https://doi.org/10.1016/j.neuron.2006.11.010>
- Otto, A. R., Gershman, S. J., Markman, A. B., & Daw, N. D. (2013). The curse of planning: Cognitive load reduces model-based choice in humans. *Psychological Science*, 24(5), 751–761. <https://doi.org/10.1177/0956797612463080>
- Pushkarskaya, H., Smithson, M., Joseph, J. E., Corbly, C., & Levy, I. (2015). Neural correlates of decision-making under ambiguity and conflict. *Frontiers in Behavioral Neuroscience*, 9, 325. <https://doi.org/10.3389/fnbeh.2015.00325>
- Rangel, A., Camerer, C., & Montague, P. R. (2008). A framework for studying the neurobiology of value-based decision making. *Nature Reviews Neuroscience*, 9(7), 545–556. <https://doi.org/10.1038/nrn2357>

Tom, S. M., Fox, C. R., Trepel, C., & Poldrack, R. A. (2007). The neural basis of loss aversion in decision-making under risk. *Science*, 315(5811), 515–518.

<https://doi.org/10.1126/science.1134239>

Tversky, A., & Kahneman, D. (1991). Loss aversion in riskless choice: A reference-dependent model. *The Quarterly Journal of Economics*, 106(4), 1039–1061.

<https://doi.org/10.2307/2937956>

Yin, H. H., & Knowlton, B. J. (2006). The role of the basal ganglia in habit formation. *Nature reviews. Neuroscience*, 7(6), 464–476. <https://doi.org/10.1038/nrn1919>